

Final Project: Digital Transformation Strategy for a Mid-sized Retail Chain

Instructions: Use this template to draft the project deliverables. You may modify the template to include more information.

Step I: Current system assessment

1. System architecture overview

The retailer operates a fragmented, largely on-premises technology environment. Point-of-sale (POS) terminals run on Windows CE with slow transaction processing (often >5 seconds) and outdated 2009 card reader drivers. The e-commerce platform is Magento 1.9 running on a separate MySQL database, with poor web performance (cumulative layout shift >0.25; page load >2 seconds on 4G). Data is stored in on-premises SQL databases at each store, with nightly batch updates sent to headquarters, resulting in fragmented data and inconsistent customer profiles. Inventory management is largely manual, using Excel spreadsheets, which causes stockouts and delayed reordering. Overall, the architecture lacks real-time integration, centralized data, and the scalability needed to support expansion to 100+ stores.

2. Pain-points matrix

Pain point	Impact	Root cause
Checkout delays	Reduced conversion rates	Slow POS processing on Windows CE
Stockouts and delayed restocking	Lost sales, poor customer experience, inefficient inventory turnover	Manual Excel-based inventory processes and lack of real-time inventory visibility
Fragmented marketing and customer data	Ineffective targeted campaigns, inconsistent customer experience	Split email lists by channel and inconsistent customer data across Magento and store databases
Slow e-commerce performance	High bounce rates, abandoned carts, suboptimal online experience	Magento 1.9 on separate MySQL database with poor page load and layout stability
Lack of omnichannel capabilities	Missed BOPIS and unified cart opportunities, limited scalability	Siloed systems, batch data updates, no unified commerce platform

3. Key inefficiencies

The current architecture restricts agility and scalability in several ways:

- Manual CSV imports and duplicate data entry create operational delays and errors.
- Customer data is inconsistent across Magento and store databases, preventing personalization.
- Slow POS and web interfaces reduce customer satisfaction and conversion.
- No real-time inventory visibility causes stockouts and missed omnichannel sales.
- On-premises, per-store databases and nightly batch updates make it difficult to scale to 100+ stores.
- Legacy POS and e-commerce platforms lack API-first integration, limiting automation and future innovation.

Step II: Stakeholder requirements

1. Stakeholder identification

Stakeholder	Interests
Customers	Fast purchases, unified cart, real-time stock visibility
Store Staff	Reliable POS with transactions ≤5 seconds; inventory visibility for quick lookups
Management	Data-driven insights via a single customer view; margin analytics; scalable growth
IT Team	Maintainability, security, compliance (API-first, GDPR); minimal on-premises footprint

2. Requirements

Document at least three functional and two non-functional requirements, justifying their alignment with stakeholder needs and business pain points.

Functional requirements:

ID	Requirement	Justification
FR01	Process in-store sales in $\leq 5s$	Addresses checkout delays for customers and staff
FR02	Real-time inventory decrement on any channel	Prevents stockouts, supports omnichannel accuracy, and enables BOPIS
FR03	360° customer profile exposed via API	Enables personalized marketing, unified customer view, and better management insights
FR04	Support buy online pickup in-store (BOPIS)	Delivers seamless omnichannel experience and customer convenience

Non-functional requirements:

ID	Requirement	Justification
NFR01	System availability $\geq 99.9\%$	Ensures reliable operations for all stakeholders
NFR02	Page load < 2 seconds on 4G	Improves online experience, reduces bounce rate, and increases conversions
NFR03	Scale to 3× holiday traffic without downtime	Supports peak demand and future expansion to 100+ stores
NFR04	API-first and GDPR-compliant security	Aligns with IT team priorities for maintainability, security, and compliance

Step III: Alternative solution evaluation

1. Comparative analysis

Weights: Functional fit 40%, scalability 20%, total cost of ownership 20%, implementation risk 10%, vendor viability 10%.

Cost and risk are normalized to a 10-point scale where higher = better.

Criteria	Weight	Salesforce Commerce Cloud + POS	Shopify Plus + Square POS	Custom Microservices
Functional fit	40%	9/10 → 3.6	8/10 → 3.2	9/10 → 3.6
Scalability	20%	10/10 → 2.0	7/10 → 1.4	10/10 → 2.0
Cost (Year 1)	20%	\$150k → 6/10 → 1.2	\$50k → 10/10 → 2.0	\$300k → 3/10 → 0.6
Implementation risk	10%	Risk 15 → 7/10 → 0.7	Risk 10 → 9/10 → 0.9	Risk 25 → 4/10 → 0.4
Vendor viability	10%	10/10 → 1.0	9/10 → 0.9	6/10 → 0.6
Total score	100%	8.5	8.4	7.2

2. Recommended solution

Salesforce Commerce Cloud + POS (Tableau Retail) is recommended. It achieves the highest weighted score (8.5), offers strong functional fit and scalability, and can support the retailer's expansion to 100+ stores. Its 6-month time to value fits within the 9-month timeline. It directly addresses stakeholder requirements: fast POS transactions, real-time inventory, 360° customer profiles, BOPIS, and high availability. While Shopify Plus is cheaper and faster to deploy, its scalability (7/10) and omnichannel depth may become limiting at 100+ stores. Custom microservices offer high customization but are costly (\$300k) and risky (12-month timeline).

3. Trade-offs

- **Cost vs. scalability:** Salesforce costs more upfront (\$150k vs. \$50k for Shopify) but provides superior scalability and enterprise-grade omnichannel capabilities.
- **Speed vs. customization:** Shopify is faster to deploy (3 months vs. 6 months), but Salesforce offers a stronger long-term platform for growth.
- **Risk vs. control:** Custom microservices offer maximum control but carry high implementation risk, longer time to value, and negative NPV until year 4.

Step IV: Feasibility and risk analysis

1. Feasibility analysis

Technical feasibility:

Cloud SaaS solutions integrate via REST APIs with minimal on-premises footprint. Salesforce supports API-first architecture, real-time inventory synchronization, and unified commerce across POS and e-commerce. Data migration from legacy SQL and Magento 1.9 is technically achievable with ETL tools and phased rollout.

Economic feasibility:

Salesforce has a positive net present value (NPV) over 5 years. Although Year 1 cost is \$150k, reduced stockouts, faster checkout, and improved conversion rates are expected to deliver ROI. Shopify is cheaper but less scalable; custom microservices have negative NPV until year 4.

Operational feasibility:

Staff upskilling is feasible within 8 weeks. A 9-month implementation aligns with seasonal sales cycles. Store staff can adapt to a modern POS, and IT can manage a cloud-based platform with minimal on-premises maintenance.

2. Risk analysis

Risk	Probability	Impact	Risk Score	Mitigation
Data migration loss	3	4	12	Two dry runs, checksum validation, rollback plan
Vendor outage	2	5	10	SLA 99.9%, offline POS mode, multi-region failover
User adoption resistance	3	3	9	Training, phased rollout, store champions
Integration complexity	3	4	12	API-first design, middleware, pilot in 5 stores

3. Mitigation effectiveness

Data migration dry runs and checksum validation ensure data integrity before go-live. Vendor outage mitigations SLA, offline POS mode, and failover keep stores operational during disruptions. Training and phased rollout reduce adoption resistance. API-first integration and a pilot reduce complexity. Together, these mitigations keep the project on track within the 9-month timeline

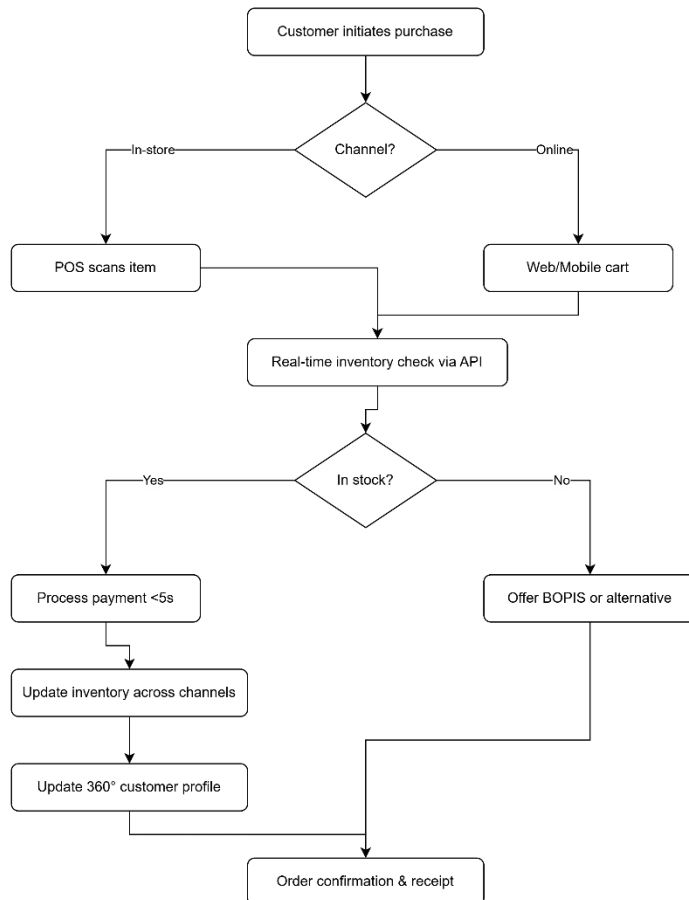
Step V: Visualizations and recommendations

1. Results and visualizations

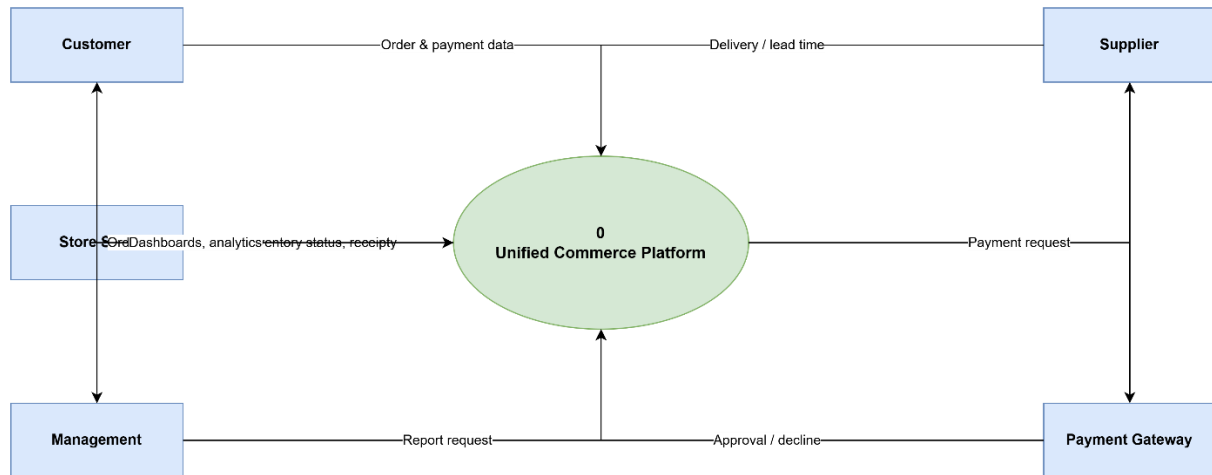
The proposed Salesforce solution addresses key pain points:

- **Checkout delays:** Modern POS processes transactions in ≤ 5 seconds.
- **Stockouts:** Real-time inventory decrement and automated reordering.
- **Marketing fragmentation:** 360° customer profile enables personalized campaigns.
- **Online experience:** Page load < 2 seconds, unified cart, and BOPIS.
- **Scalability:** Cloud platform scales to 100+ stores and 3× holiday traffic.

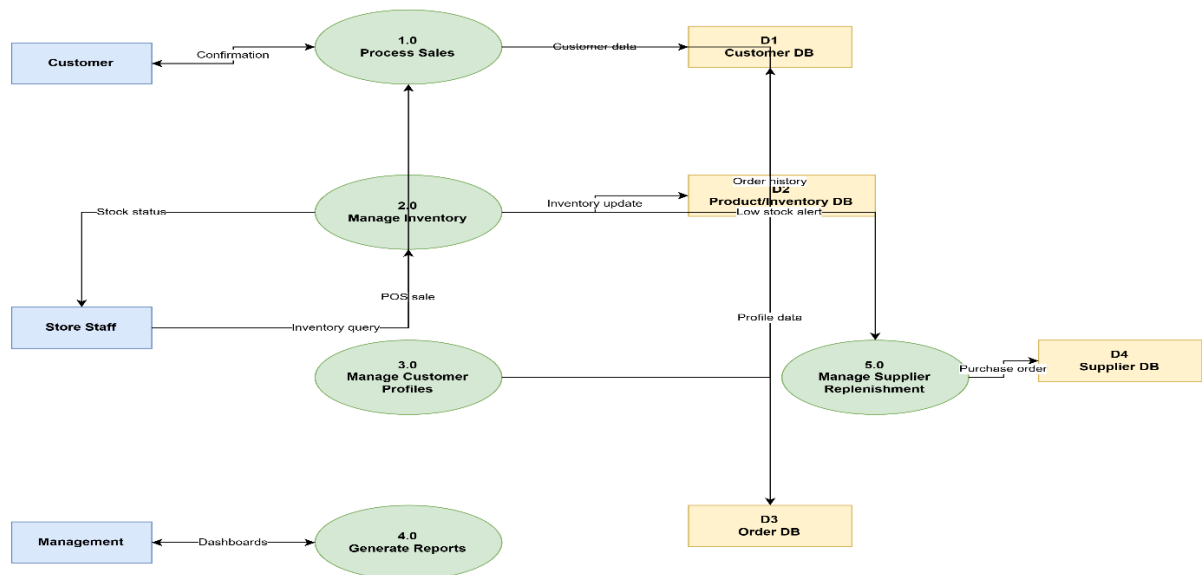
Visual artefact



- FIGURE 1. Process flow for omnichannel checkout, illustrating improved transaction speed and inventory visibility.



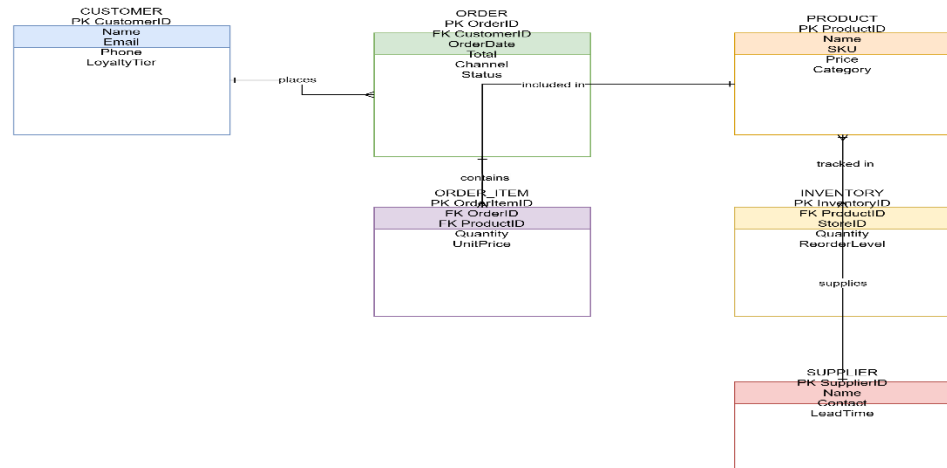
Context Diagram (Level 0 DFD) — Unified Commerce Platform



Level 0 DFD — Unified Commerce Platform Decomposition

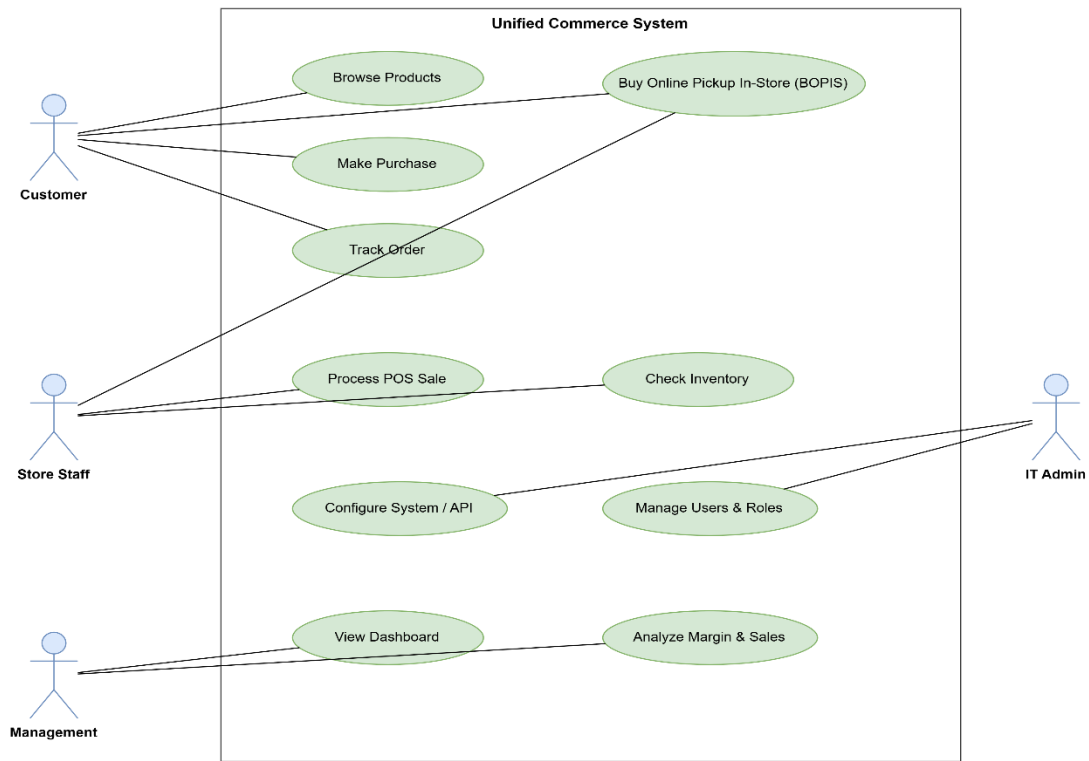
- FIGURES 2: Context and level 0 DFDs showing system interactions.

Text



Customer, Order, Product, Inventory, and Supplier Entities

- FIGURE 3: ERD detailing customer, order, product, inventory, and supplier entities.



UML Use Case Diagram — Unified Commerce System

- FIGURE 3: UML diagrams such as use case diagrams, sequence diagrams for checkout, and activity diagrams for restock.

2. Implementation roadmap

Key milestones for implementation within nine months

Milestone	Timeline	Description
Discovery & vendor selection	Month 1	Finalize requirements, select Salesforce, kick off project
Data migration planning & API design	Month 2	Design integration, map legacy data, set up cloud environment
POS pilot in 5 stores	Month 3	Test POS, inventory sync, and checkout speed
E-commerce integration	Month 4	Connect Magento replacement to unified commerce platform
Data migration dry run 1	Month 5	Validate data integrity, checksum, rollback procedures
Data migration dry run 2 & BOPIS pilot	Month 6	Second dry run; pilot BOPIS in select stores
Full POS rollout to 50 stores	Month 7	Deploy POS chain-wide; begin staff training
Omnichannel features full rollout	Month 8	Unified cart, real-time inventory, 360° customer view
Performance tuning & security audit	Month 9	Scale testing, GDPR compliance, hypercare, expansion readiness